Jacomo Corrieri

jacorrieri@gmail.com | linkedin.com/in/jacomo-corrieri | jcorrieri.github.io/Portfolio | US Citizen

EDUCATION

University of Florida, *Gainesville, FL* *Expected May 2027*
Master of Science (M.S.) in Computer Science
GPA: 3.83/4.0
Relevant Coursework: Algorithms, Computer Vision, Distributed Systems, CUDA Programming

University of North Florida, *Jacksonville, FL* *May 2025*
Bachelor of Science (B.S.) in Computer Science, Minors in Mathematics and International Business
GPA: 3.99/4.0 | *Summa Cum Laude* | Hicks Honors College
Relevant Coursework: Algorithms, Data Structures, Systems Programming, Vector Calculus

WORK EXPERIENCE

Florida Blue, *Jacksonville, FL* *May 2026 – Present*
Information Technology Intern – Software Engineering
- Aggregate relational data across **8+ tables** and **2 databases** for monitoring and standardization using SQL and Java.
- Create and improve **28 API endpoints** in Spring Boot and FastAPI, testing latency and correctness with PyTest.
- Build a React dashboard to visualize migration failure telemetry, accelerating root-cause triage by engineering teams.
- Collaborate across teams to integrate, test, and deploy application services with Jenkins and OpenShift.

UFIT Research Computing, *Gainesville, FL* *Jan 2026 – May 2026*
Research Computing Support Intern
- Built and deployed a Python CLI and Django REST API to analyze metrics for **800M+ software usage records.**
- Triaged, debugged, and resolved **30+** production HPC issues for HiPerGator users, including an **8-month backlog** ticket.
- Increased deprecation check speed by **10–30×** by caching read-heavy filesystem data in PostgreSQL tables.
- Automated nightly Slurm cron jobs for data ingestion and containerized production applications with Podman.

RESEARCH

IEEE ICTAI 2026 - *Scalable Multi-robot Coverage with Connectivity Maintenance Via Deep Reinforcement Learning*
J. Corrieri, I. Kulbaka, A. Ghosh, A. Dutta
- Simulated up to 50 autonomous robots in Python and Gymnasium, achieving **5x more scalability** than prior work.
- Developed a decentralized CNN model with PyTorch, resulting in **15% greater area coverage** compared to the baseline.

PROJECTS

Tiny DNS Cache *Sep 2026 – Present*
- Build a multithreaded C++20 DNS cache using Unix sockets, thread pooling, and a bounded producer-consumer queue.
- Design a thread-safe templated cache with a shared mutex, plus a RAII SQLite wrapper and CTest (CMake) tests.
- Develop record lookup with recursive CTEs for CNAME/NS chains, zero-copy views, and binary IPv4/IPv6 decoding.

MoTorch Neural Network Library *Apr 2026 – Jul 2026*
- Created a PyTorch-inspired deep learning framework with neural network modules, optimizers, and loss functions.
- Implemented a reverse-mode autodiff engine for tensors with a dynamic computational graph for backpropagation.
- Automated CI and testing with GitHub Actions and PyTest, benchmarking models with up to **32.6 million parameters**.

UF Student Marketplace *Feb 2026 – Jul 2026*
- Engineered a full-stack Angular/TypeScript and Go application using WebSockets for real-time messaging.
- Replaced database blobs with scalable cloud-based object storage by integrating AWS S3 with a Gin REST API.
- Designed and indexed relational schemas with GORM and SQLite for listings, messaging, and user workflows.

SKILLS

Languages: Python, C++, Go, Java, TypeScript, SQL
Libraries: FastAPI, Django, Gin, Spring Boot, React, CUDA, PyTorch, NumPy, Gymnasium
Infra & Tools: Linux, Slurm, Git, GitHub Actions, Podman, PostgreSQL, SQLite, AWS S3, CMake, Jenkins, OpenShift